StepWise — Response to Review-3 Comments

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We thank the panel for the comments. Each is answered below with what we did, what we found and where it appears in the revised paper. Every number is regenerated by `python -m app.ml.validate` and stored in `backend/ml_models/reports/validation_*.json`. Three comments need data that only a classroom can provide. For those we report the closest evidence available now, say plainly what is still open, and give a ready-to-run protocol (`docs/REAL_WORLD_VALIDATION.md`).

C1. Only 56.7% exact expression accuracy may limit downstream diagnosis

Agreed, and confirmed.

- *Why 56.7%.* The 127 test lines average 9.35 symbols, not about 4 as the Review-3 text said (that figure was an error). With 92.4% symbol accuracy, independent errors predict 54.2%. Lines of up to 5 symbols, the size of most student steps, are read exactly 86% of the time. Of the failed lines, 26 are segmentation errors (touching symbols) and 29 are substitutions.
- *Improvement.* Syntax-constrained decoding (Viterbi over the top-3 classes of each symbol, well-formed lines only, never using arithmetic truth) raises exact lines to **61.4%**.
- *Cost to diagnosis.* On 420 synthetic solutions written in real CROHME test-writer handwriting, diagnosis accuracy falls from **99.3%** on the true transcription to **35.0%** on the recognizer's reading, and 57% of correct work is marked wrong.
- *Mitigation in the app.* Confirm-before-diagnose: symbols below confidence $\tau = 0.98$ (chosen on separate validation pages) are highlighted, and no diagnosis is given until the student checks them. Accuracy returns to **93.8%** and correct work marked wrong falls to 5%, at a cost of about 7 of 14 symbols checked per page (trade-off in Fig. 2). StepWise is therefore semi-automatic until the reader improves. The study also exposed two crash bugs on misread input, both now fixed.

Paper: Sections IV-B, IV-C, V-B; Table VII; Fig. 2.

C2. The misconception classifier is trained on synthetic solutions; validate on real student work

Done with real student errors; real handwritten work still to collect.

- We evaluated M2 on **CogTutorBugs** (CC BY 4.0): steps on which 95 Algebra I students made an error caught by the Cognitive Tutor. SymPy separates 2,120 mathematically wrong steps from 3,100 valid but unhelpful ones. The data are typed, so this tests diagnosis alone.
- *Detection transfers:* 94–96% of real wrong steps are flagged, 99.3% of valid steps accepted (old rule-based system: 10.8%).
- *Naming the category does not, at first:* the Review-3 model was right for **46.3%** of errors on held-out problem templates, below the 53.2% majority class (macro-F1 0.279, against 0.994 on synthetic data).
- *Why, and v2.* Error analysis on a separate 30% of templates showed that students slip while simplifying one side in place ($-75x = 4 - 54 \rightarrow -75x = 58$), which our generator never produced. Nine new features and matching generators (M2 v2, still trained only on synthetic data) give **52.1%**, macro-F1 **0.460** on the held-out templates. In cross-validation, adding about 160 real labelled errors lifts accuracy to **77%**, so labelled real data is the remaining lever.
- *Next:* Study A in `REAL_WORLD_VALIDATION.md`: 200 photographed pages from at least 30 students, double-annotated (Cohen's $\kappa \geq 0.70$), evaluated by `evaluation/real_world_eval.py`.

Paper: Sections IV-E, V-E; Tables IX–X.

C3. Evaluate on actual smartphone photographs and diverse handwriting styles

Partly done; real phone photos are part of Study A.

- *Simulated capture* of the 420 pages under 12 conditions (paper and pen, notebook rules, uneven lighting, cast

shadow, tilt, blur, noise and JPEG, low resolution, three combined severities). It exposed a design flaw: a single global threshold reads shadows and rules as ink. A new photo-robust page binarization reduces CER on ruled paper from 0.90 to 0.16, under uneven lighting from 0.74 to 0.10 and under a cast shadow from 0.80 to 0.12, with clean images unchanged. Combined moderate or severe capture still defeats the reader.
- *Writer populations.* On MNIST test digits (pen on paper, never used for training) the recognizer reads 86.3% of digits by about 250 high-school students and 85.0% by 48 Census employees, against 91.0% on CROHME digits. The main failure is a slanted '1' read as '/'.

Paper: Sections IV-D, V-E; Table VIII.

C4. Clarify the mapping between ASSISTments skills and the eight tutoring skills

Done. The new skill-mapping table gives, per StepWise skill: the ASSISTments skills, match quality (direct: S1, S3, S4; close: S5, S6; partial: S2; approximate: S7; none: S8, which keeps simulated parameters), data volume, the deployed BKT parameters and test AUC on those skills. We also clarify that the AUC 0.815 benchmark uses all 110 ASSISTments skills; the mapping only transfers BKT parameters. Two findings: several parameters sit on the EM bounds, and BKT predicts S7 barely above chance (AUC 0.54), so its S7 mastery estimate is weak. *Paper:* Section IV-F; Table XII.

C5. Validate adaptive practice through a real classroom study, not simulation alone

Agreed; protocol ready, trial not yet run.

- *Robustness now.* The deployed learner model chose items for learners from five simulators. Adaptive practice beats random with the original simulator (post-test $d = 0.37$), with learners calibrated on real students ($d = 0.50$) and with learners who differ in learning rate ($d = 0.29$). It shrinks when learners forget ($d = 0.17$) and vanishes when these are combined ($d = 0.04$), because the policy stops practising skills that BKT, which has no forgetting, calls mastered. Simulation therefore cannot settle the question, as the panel noted.
- *Classroom trial (Study B):* randomized at student level, stratified by class and pre-test, comparing adaptive with random practice from the same item bank: pre-test, four sessions of 10 items, post-test and a two-week delayed post-test. The primary analysis is ANCOVA. Detecting $d = 0.3/0.4/0.5$ needs 132/75/49 students per arm, so a two-class first run is a pilot that estimates the effect size and feasibility.

Paper: Sections IV-G, V-E; Tables XIII–XIV.

Other corrections

- The test-suite table now lists the real per-module counts (171 cases in 14 modules); the Review-3 table did not add up to its total.
- The mistake classifier is now v2 (42 features); its synthetic macro-F1 is 0.992 (was 0.994).
- The limitations, conclusion and future work were rewritten to match the new evidence. StepWise is presented as ready for a supervised classroom pilot, not for unsupervised use.